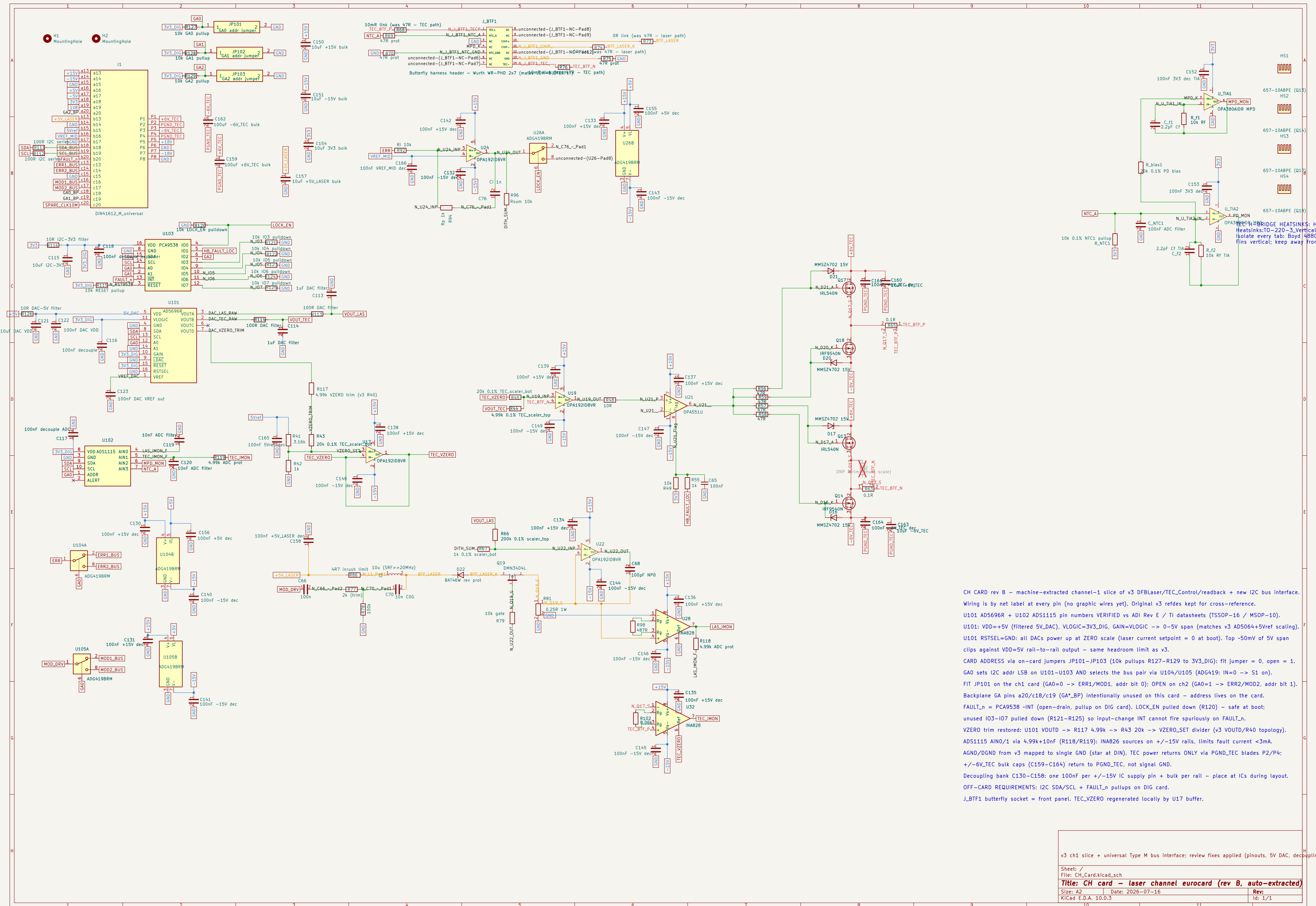




CH CARD rev B – machine–extracted channel–1 slice of v3 DFBLaser/TEC_Control/readback + new I2C bus interface. Wiring is by net label at every pin (no graphic wires yet). Original v3 refdes kept for cross-reference.

U101 AD5696R + U102 ADS1115 pin numbers VERIFIED vs ADI Rev E / TI datasheets (TSSOP–16 / MSOP–10).

U101: VDD=+5V (filtered 5V_DAC), VLOGIC=3V3_DIG, GAIN=VLOGIC -> 0–5V span (matches v3 AD5064+5Vref scaling).

U101 RSTSEL=GND: all DACs power up at ZERO scale (laser current setpoint = 0 at boot). Top –50mV of 5V span clips against VDD=5V rail–to–rail output – same headroom limit as v3.

CARD ADDRESS via on–card jumpers JP101–JP103 (10k pullups R127–R129 to 3V3_DIG): fit jumper = 0, open = 1.

GA0 sets I2C addr LSB on U101–U103 AND selects the bus pair via U104/U105 (ADG419: IN=0 -> S1 on).

FIT JP101 on the ch1 card (GA0=0 -> ERR1/MOD1, addr bit 0); OPEN on ch2 (GA0=1 -> ERR2/MOD2, addr bit 1).

Backplane GA pins a20/c18/c19 (GA*_BP) intentionally unused on this card – address lives on the card.

FAULT_n = PCA9538 -INT (open–drain, pullup on DIG card). LOCK_EN pulled down (R120) – safe at boot; unused IO3–IO7 pulled down (R121–R125) so input–change INT cannot fire spuriously on FAULT_n.

VZERO trim restored: U101 VOUTD -> R117 4.99k -> R43 20k -> VZERO_SET divider (v3 VOUTD/R40 topology).

ADS1115 AIN0/1 via 4.99k+10nF (R118/R119): INA826 sources on +/-15V rails, limits fault current <3mA.

AGND/DGND from v3 mapped to single GND (star at DIN). TEC power returns ONLY via PGND_TEC blades P2/P4; +/-6V_TEC bulk caps (C159–C164) return to PGND_TEC, not signal GND.

Decoupling bank C130–C158: one 100nF per +/-15V IC supply pin + bulk per rail – place at ICs during layout.

OFF–CARD REQUIREMENTS: I2C SDA/SCL + FAULT_n pullups on DIG card.

J_BTf1 butterfly socket = front panel. TEC_VZERO regenerated locally by U17 buffer.